Collaboratory

# grEPI

Global Repository of Epidemiological Parameters

## User Guide

*October 2026, v10*

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# 1. Introduction

grEPI (Global Repository of Epidemiological Parameters) is a publicly accessible, curated repository of epidemiological parameter estimates for infectious diseases. It is developed by the WHO Hub for Pandemic and Epidemic Intelligence in collaboration with the GREP Community of Practice, a global multidisciplinary network of over 100 epidemiologists, modelers, public health practitioners, data scientists, and IT professionals.

The repository brings together parameter estimates extracted from peer-reviewed literature. It provides a standardized way to capture, present, and access this data for use in infectious disease modelling, surveillance, public health decision-making, and research synthesis. Each parameter estimate is versioned, meaning that when a record is updated, the previous version is retained and remains viewable, so the history of an estimate is preserved alongside the current value.

grEPI is currently in its proof-of-concept phase, covering parameter estimates for a prioritized set of infectious diseases including Ebola disease, Lassa fever, Marburg disease, Measles, Mpox, Severe Acute Respiratory Syndrome (SARS), and Zika virus disease.

The repository currently contains estimates across the following parameter types: delay distributions (incubation period, serial interval, generation interval, infectious period, latent period), severity measures (case fatality ratio, case hospitalization risk, proportion of hospitalizations requiring critical care, proportion of infections hospitalized), transmissibility indicators (basic, effective, and maximum reproduction numbers, growth rate, doubling time, overdispersion), attack rates (primary and secondary), seroprevalence (by assay type, including IgG, IgM, PRNT, IFA, MIA, and healthcare-associated infections), mutation rates (evolutionary rate, mutation rate, substitution rate), and relative contribution.

You can access grEPI data in two ways: through the web application, or programmatically via the grEPI public API. Almost all grEPI content is available without an account, but full access to all reference tables and role-based features requires signing in with a WHO Collaboratory account, which anyone can create at https://collaboratory.who.int.

The links below provide access to grEPI and related resources.

### Key Links

| Resource | URL |
|---|---|
| grEPI web application | https://collaboratory.who.int/epidemiologicalparameters/repository/ |
| grEPI website | https://who-collaboratory.github.io/collaboratory-grepi-web/ |
| grEPI API user guide | https://who-collaboratory.github.io/collaboratory-grepi-web/assets/files/grEPI_API_Documentation.pdf |
| grEPI public API endpoints(Swagger UI) | https://collaboratory.who.int/grepi/api/swagger/index.html |
| Epidemiological Parameters Community of Practice website | https://who-collaboratory.github.io/collaboratory-epiparameter-community/ |
| Epidemiological Parameters Community of | https://collaboratory.who.int/about/communities/epidemiological-parameters/ |

| | |
|---|---|
| Practice Collaboratory website | |

## 2. Navigating the Web Application

The grEPI web application opens to a homepage that provides an overview of the repository. The application is organized into modules, which are accessible from the left sidebar. Each parameter estimate in the repository links to a source publication (literature) and is classified using controlled vocabularies from a set of reference data tables. These sections reflect that structure:

**The Homepage** is the default landing view when you open the grEPI web application. It provides an overview of the repository's current contents and an entry point into the data. The page has several sections:

- *Welcome tile provides* a short introduction to grEPI, including its purpose and the current proof-of-concept scope. A "Find out more" link navigates to the grEPI web page.
- *Getting started tile* provides quick access to grEPI web application User Guide and API documentation.
- *Help grEPI grow tile* encourages users to contribute to grEPI byproviding feedback, reporting issues, suggesting improvements, and sharing epidemiological parameter estimates extracted from the literature or generated through their own work. A contact link provides a direct way to reach out to the Collaboratory team.
- *Interactive live counts by disease tile* lists every disease currently covered in grEPI with a live count of parameter estimates available for that disease. The counts update as new estimates are added to the repository, so the figures shown always reflect the current state of the data. Each disease name and count are links: clicking these opens the Epidemiological Parameters browse table pre-filtered for that disease. Diseases with at least one estimate added in the last 30 days will be marked with a "New" badge. The tile also includes a "Total added in 30 days" counter, which shows the total number of parameter estimates added in grEPI in the last 30 calendar days.
- *Interactive stacked bar chart with estimates grouped by disease and WHO region.* shows the number of parameter estimates per disease, additionally grouped by the WHO region. The legend on the right identifies each WHO region by color. Hovering over a bar segment (stack) shows a tooltip with the disease name, the region, and the exact count for that segment. Clicking a bar segment (stck) opens the Epidemiological Parameters browse table pre-filtered by selected disease and WHO region. As with the disease counts, the chart reflects the live state of the repository.
- *Interactive pie chart with estimates grouped by Parameter Class* shows the distribution of parameter estimates across parameter classes (i.e., , Delays, Severity, Population Immunity, and Transmissibility). Hovering over a chart segment displays a tooltip showing the class name, count, and percentage of all estimates. Clicking a chart segment opens the Epidemiological Parameters browse table pre-filtered to the selected parameter class.
- *Interactive stacked bar chart with estimates grouped by Disease and Parameter Class* shows the number of parameter estimates per disease, additionally grouped by

parameter class. The legend on the right identifies each class by color. Hovering over a bar segment shows a tooltip with the disease name, the parameter class, and the exact count for that segment. Clicking a bar segment opens the Epidemiological Parameters browse table pre-filtered to that disease and parameter class. As with the other Home page visualizations, the chart reflects the live state of the repository.




*Figure 1. grEPI web application homepage*

**Epidemiological Parameters** (Section 3) contains the core dataset: extracted parameter estimates from the literature. Each row represents a single estimate (e.g., an incubation period for Mpox from a specific study), with details on the study population, geographic context, statistical methods, and source citation.

**Literature** (Section 4) contains details on the publications from which estimates have been extracted. Each row represents a literature entry (e.g., journal article, grey literature source). A single literature entry may be the source for one to multiple parameter estimates.

**Reference Data** (Section 5) contains the controlled vocabularies and classification tables that structure the repository. These include disease definitions (linked to ICD codes), pathogen taxonomy (species, genus, family), epidemiological parameter categorizations, delay event definitions, estimate units, and estimate value types. There are currently 35 reference tables. These tables define the valid values used across the Epidemiological Parameters and Literature sections. Without signing in, only two reference tables are visible: “Diseases caused by pathogens (strains, clades, serotypes)” and “Pathogen taxonomy.” To access the full set of reference tables, sign in with your existing WHO Collaboratory account or create a new one.

Additional interface elements:

**Breadcrumb bar.** Displayed at the top of each page, it shows your current location within the application (e.g., Home > Reference Data > Diseases).

**Top-right icons in the header.** The diamond-shaped icon links to the grEPI API public endpoints (Swagger UI). The Sign In button allows you to create or log in with a Collaboratory account.

**grEPI logo in the header.** The grEPI logo icon allows you to always return to the homepage.

**Top-left icon in the header.** The hamburger menu icon allows you to collapse and expand the sidebar menu.

**Help icons.** In the parameter detail view, circled question mark icons appear next to field labels. Hover over these for a brief description of what the field means.

# 3. Epidemiological Parameters

## 3.1 The Browse Table

The Epidemiological Parameters module opens to a paginated table of all parameter estimates in the repository. By default, the table shows a subset of the available columns. The full dataset contains over 100 fields. Additional columns can be made visible using the Select button, which opens a dropdown with a column search, "Select All", and "Remove All" options. Columns that are currently visible in the table are marked as checked in the Select dropdown list.

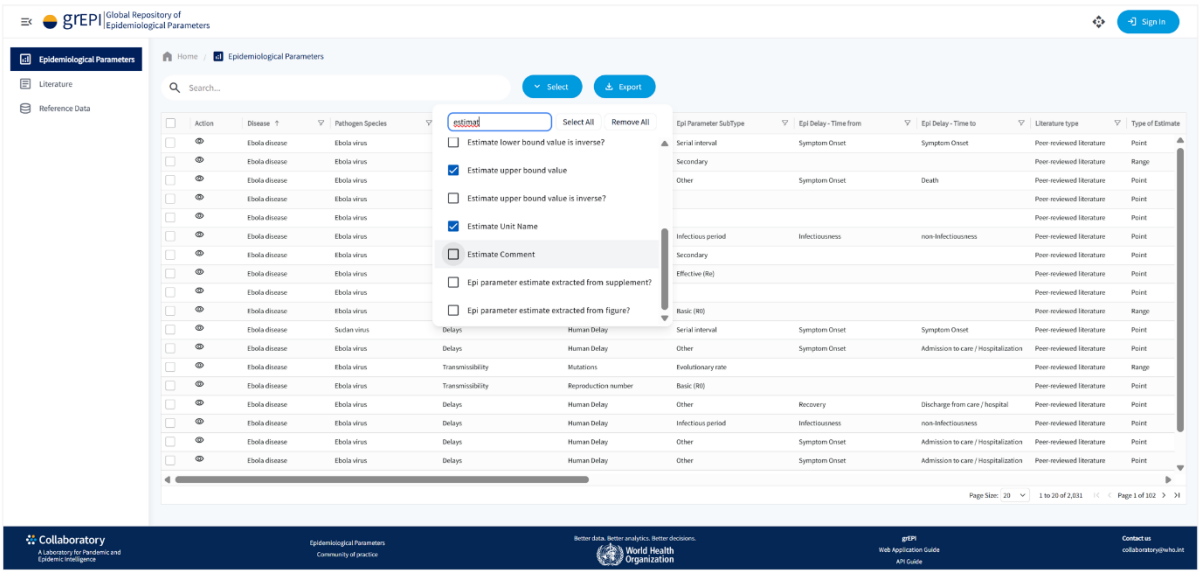



*Figure 2. The Epidemiological Parameters browse table, showing the default column layout with expanded and filtered Select menu*

**Default visible columns:** Disease, Pathogen Species, Parameter Class, Parameter Type, Parameter Subtype, Delay Start Event, Delay End Event, Literature type, Type of Estimate, Estimate Value Type, Estimate Point Value, Estimate Lower Bound value, Estimate Upper Bound value, Estimate Unit, Countries (Population), Population study (start year), Population study (end year), WHO Region, Created date, Version number.

The table below lists the remaining columns available via Select, grouped thematically for reference. Note that, in the application, all columns appear in a single flat list within the Select dropdown.

| Field Type | Fields |
|---|---|
| Identifiers | grEPI_ID |
| Literature | Literature Unique Identifier Type, Literature Unique Identifier, Article Title, Article Authors, Article Published in, Article Publication year, Article Label, Literature Volume, Literature Issue, Article page first, Article Page last |

| | |
|---|---|
| Disease classification | Disease Code ICD, Disease Subchapter Name ICD, Disease Chapter Name ICD, Disease Chapter Code ICD, Disease Spread Type, Disease Pathogen Comment |
| Pathogen taxonomy | Pathogen Species Code ICD, Pathogen Strain, Pathogen Strain Code ICD, Pathogen Substrain, Pathogen Substrain Code ICD, Pathogen Genus, Pathogen Genus Code ICD, Pathogen Family, Pathogen Family Code ICD, Pathogen Transmission |
| Estimate categorization | Estimate category comment |
| Estimate values | Estimate point value is inverse?, Estimate Lower bound value is inverse?, Estimate Upper bound value is inverse?, Estimate Comment |
| Uncertainty | Uncertainty value type, Uncertainty Single Value, Uncertainty single value is inverse?, Uncertainty range value type, Uncertainty range - lower bound value, Uncertainty range - lower bound value is inverse?, Uncertainty range - upper bound value, Uncertainty range - upper bound value is inverse?, Uncertainty Unit |
| Extraction flags | Epi parameter estimate extracted from supplement?, Epi parameter estimate extracted from figure? |
| Probability distribution | Distribution type, Distribution comment, plus for each of Parameters 1-3: value available?, value, value type, uncertainty value available?, uncertainty single value, uncertainty range lower/upper bound value, uncertainty value type |
| Methods | Statistical approach, Method to estimate reproduction number, Inference method, Inference comment, Is data disaggregated?, Data dsaggregated by, Is data bias adjusted?, Is data discretised?, Is data right censored?, Is data left censored?, Is data interval censored?, Is data right truncated?, Is data left truncated? |
| Percentage calculation | Percent Calculation, Percent Adjusted by, Numerator, Numerator Definition, Denominator, Denominator Definition |
| Case classification | Case classification - confirmed, Case classification - probable, Case classification - possible/suspected, Case classification - unspecified |
| Population | Population sample size, Population age min/max years, Population age min/max months, Population sex, Population group, Population setting, Population data - moment value, Continent, Population location, Population comment |
| Data source | Data extracted by Name, Data extracted by WebSite, Data source name, Data source location, Data imported from Name, Data |

| | |
|---|---|
| | imported from URL, Data imported from DOI, Data imported on, Data imported - local unique identifier, Import comment. |

To reset your current custom view of columns and return to the default view, click Reset button.

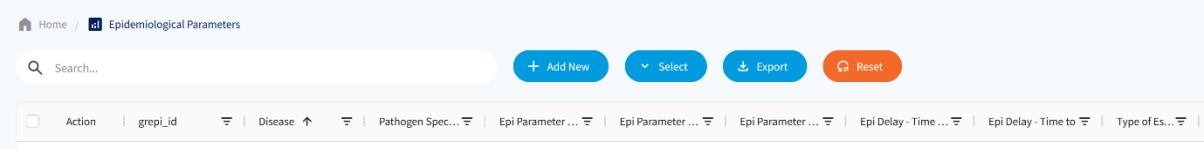



*Figure 3. The Epidemiological Parameters browse table, showing the Reset button*

**Searching.** The search bar at the top of the table performs a multi-term search across all fields. You can enter multiple keywords separated by spaces, and the table will filter to rows that match all of those terms across any combination of fields. For example, entering "mpox incubation period germany" will return rows where the disease is Mpox, the parameter subtype is Incubation period, and the Countries (Population) field includes Germany. This makes it easy to locate specific estimates without needing to set multiple column filters.

**Filtering by column.** Each column header includes a filter icon. Click it to apply a filter to that column. The filter supports multiple operators, with the available set depending on the column type.

- *Text columns* support: Contains, Does not contain, Equals, Does not equal, Begins with, Ends with, Blank, and Not blank.
- *Numeric columns* (e.g., estimate values) support: Equals, Does not equal, Greater than, Greater than or equal to, Less than, Less than or equal to, Between, Blank, and Not blank.
- *Date columns* (e.g., Created date) support: Equals, Does not equal, Before, After, Between, Blank, and Not blank.

Multiple column filters can be active at the same time.

To clear all applied column filters in one go, click Clear button.

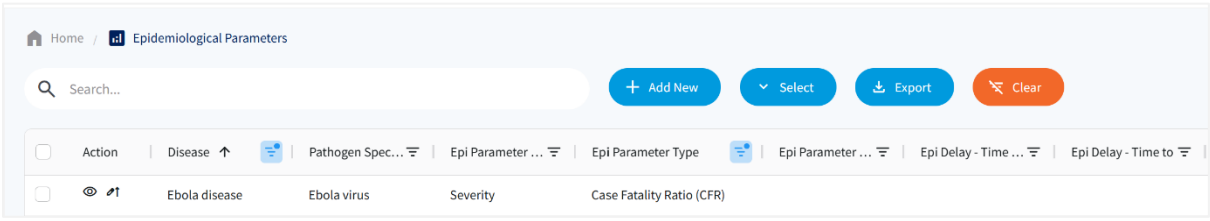



*Figure 4. The Epidemiological Parameters browse table, showing the Clear button*

**Sorting.** Click a column header to sort the table by that column. The sort direction is indicated by an arrow. Only one sort can be active at a time (sorting by a new column replaces the previous sort). The default sort is by Disease (ascending), which groups estimates for the same disease together.

**Reordering columns.** You can reorder columns by clicking on a column header and dragging it to a new position in the table.

**Pagination.** The table displays 20 rows per page by default. The Page Size dropdown offers three options: 20, 50, or 100 rows per page. Navigation controls at the bottom allow moving between pages (first, previous, next, last). The total number of records is shown (e.g., "1 to 20 of 1000").

**Copying values.** Values displayed in the browse tables can be copied directly from the grid without opening the detail view. To copy a value, click and hold the left mouse button to highlight the desired text, then right-click and select “copy” from the context menu. This functionality is available in the Epidemiological Parameters, Literature, and Reference Data browse tables.

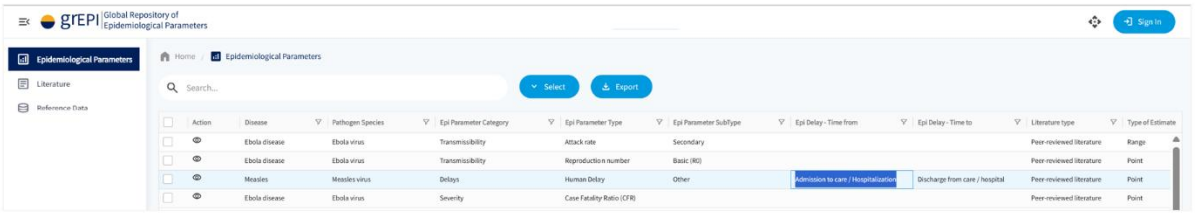



*Figure 5. The Epidemiological Parameters browse table, showing the Copy value feature*

## 3.2 The Parameter Detail View

Click the eye icon in the Action column to open the full detail page for a single parameter estimate. This page contains significantly more information than the table view, organized into the following sections.

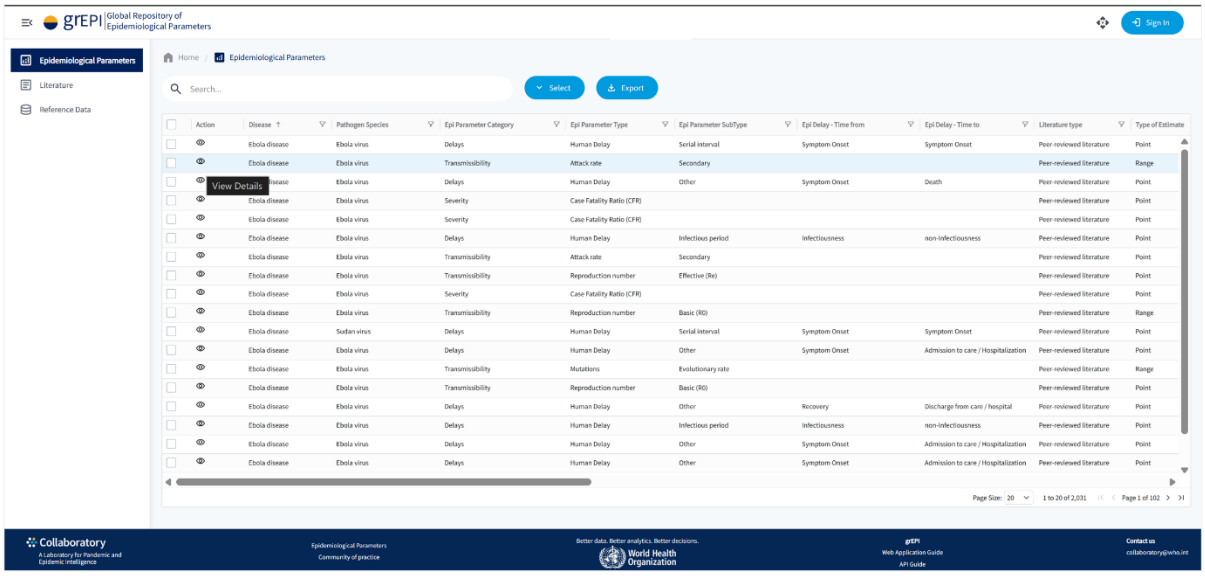



*Figure 6. The Epidemiological Parameters browse table – Action column – View Details of the record*

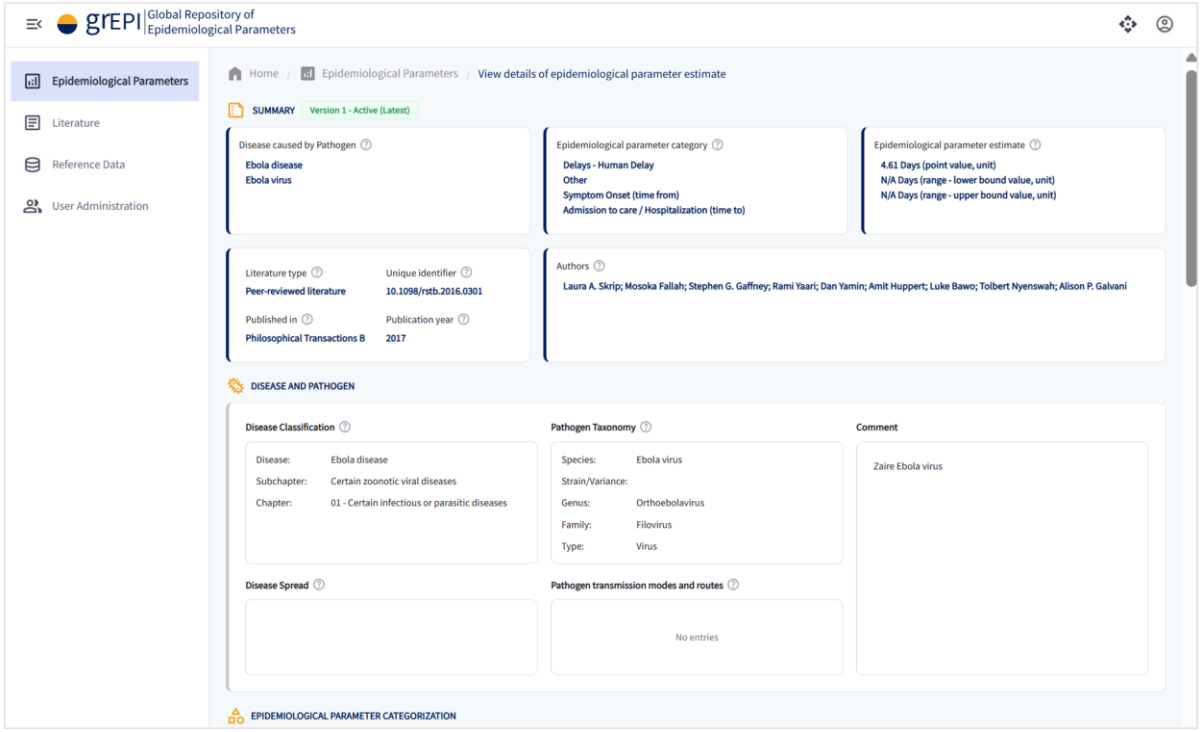



*Figure 7. The parameter detail view for a single estimate, showing the Summary cards, literature citation, and the Disease Pathogen section.*

### Summary

The top of the page shows three cards: Disease caused by Pathogen (disease name and pathogen species), Epidemiological Parameter Classification (e.g., "Severity - Case Fatality Ratio"), and the Epidemiological parameter estimate itself (point value and/or range with units). A version tag is also displayed at the top of the page. The tag shows the version number plus "Active (Latest)" when viewing the current version, or "Previous (Not current)" when viewing an earlier version. When more than one version exists, a "Compare versions" button appears next to the version tag. Below these, the Literature type, literature unique identifier, journal name, publication year and author list are displayed.

### Disease and Pathogen

Disease Classification shows the disease name, ICD subchapter, and ICD chapter. Pathogen Taxonomy shows the species, strain/substrain, genus, family, and type. Disease Spread and Pathogen Transmission are also shown (may read "Unspecified" for some entries). A Comment field captures additional context (e.g., "Zaire Ebola virus").

### Epidemiological Parameter Classification

Parameter Class is the high-level category (Delays, Severity, Transmissibility, or Population immunity); Parameter Type is the specific parameter type/subtype (e.g., Case Fatality Ratio, Human Delay - Incubation period). For delay parameters, the Epidemiological Delay sub-section shows the Delay Start Event and Delay End Event that define the interval being measured.

### Epidemiological Parameter Estimate

The Estimate card shows Value type (Mean, Median, Crude proportion, Standard deviation, Other, or Unspecified), Point value, Lower bound value, Upper bound value, Unit, and an Inverse flag. An accompanying Uncertainty card shows any reported uncertainty measures from the source study in the same structure (value type, single value, range, unit). Statistical approach shows exact value if available, otherwise shows as blank. A Comment field is available for additional notes.

### Probability Distribution

If the source study reported a fitted distribution, this section records the Distribution Type (e.g., gamma, log-normal) and up to three distribution parameters (Parameter 1, 2, 3) with their values. These typically represent that shape and scale of the distribution, or equivalent parameterizations such as mean and standard deviation, depending on the distribution type. Each parameter has corresponding Uncertainty fields (Uncertainty 1, 2, 3) with single values and ranges. Many entries will have this section empty.

### Methods

Inference Method records how the estimate was derived. Methodological criteria flags indicate whether the data was bias-adjusted, censored (right, left, or interval), discretized, or truncated (right or left). For percentage-based estimates, a Percent (%) calculation sub-section shows the Percent Calculation method (e.g., Naive), Percent Adjusted By (e.g., Age), Numerator value, Denominator value, Numerator definition, Denominator definition, and Case classification (confirmed, probable, possible/suspected, unspecified). Data disaggregation sub-section shows

whether the data is aggregated or disaggregated, and if disaggregated then disaggregation dimensions.

### Population Data

This section describes the study population from which the estimate was derived: Sample Size, Timing related to outbreak (e.g., "Post outbreak"), Setting (e.g., "Hospital based"), Population study dates (start/end year, month, day), Group (e.g., “Children”, “Pregnant women”), Age (min/max years and months), Sex, Administrative level 0 - Country/territory/area (with Name, Continent, and WHO Region), and Population location (below Administrative level 0). A Comment field provides additional location or context details.

### Literature

The source publication details: Literature type, Unique identifier, Unique identifier type, Published in (journal), Publication year, Title, Label (short citation, e.g., "Xu 2016"), Authors, and Data extracted from literature by (the organization or group that performed the extraction, with a link). Also records whether data was extracted from supplementary material or from a figure (TRUE/FALSE). For articles that are available in paper copy only volume, issue, page first and page last details are shown.

### Data Sources

Where the data was imported from: the source name, a comment on the import (e.g., "Partial import, entire dataset available at source"), the Unique local identifier from the source system, and the Data imported on date.

### Versioning

Each row in the table represents the latest version of a parameter estimate. The browse table only ever shows the current version of each record.

Previous versions are accessible from the parameter detail view (see Section 3.2) and through the version history (see Section 3.3).

At the bottom of the page, the Back button returns to the browse table.

## 3.3 Version History and Comparing Versions

When a parameter estimate has more than one version, the detail view exposes two version-related features:

**View version history.** Opens a list of all versions of the record, ordered from most recent to oldest. Each entry shows the version number, the date the version was created, the user who created it, an indicator for which version is current, and the version comment (a short note describing what changed, where provided).

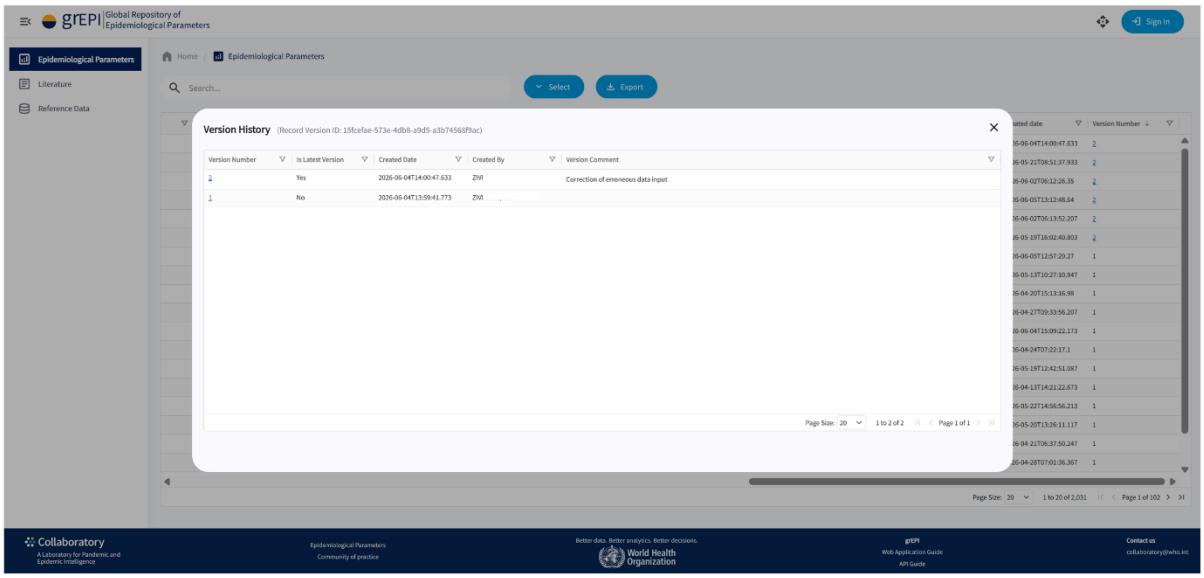



*Figure 8. The Epidemiological Parameters browse table, showing the View version history feature*

**Compare versions.** Opens a side-by-side view of two versions of the record. Fields that differ between the two versions are highlighted in green (added), yellow (modified), red (deleted), so changes to values, units, methods, or any other captured attribute are immediately visible.

Default configurations for comparing versions:

- If one is currently consulting Active (Latest) version then selection of the desired Previous version in Compare versions dropdown menu needs to be selected to show comparison

- If one is currently consulting Previous (not Active/Latest) version then that version will automatically be compared to the Active (Latest) version in the Compare versions. However, one can always adjust which version to compare with another version.

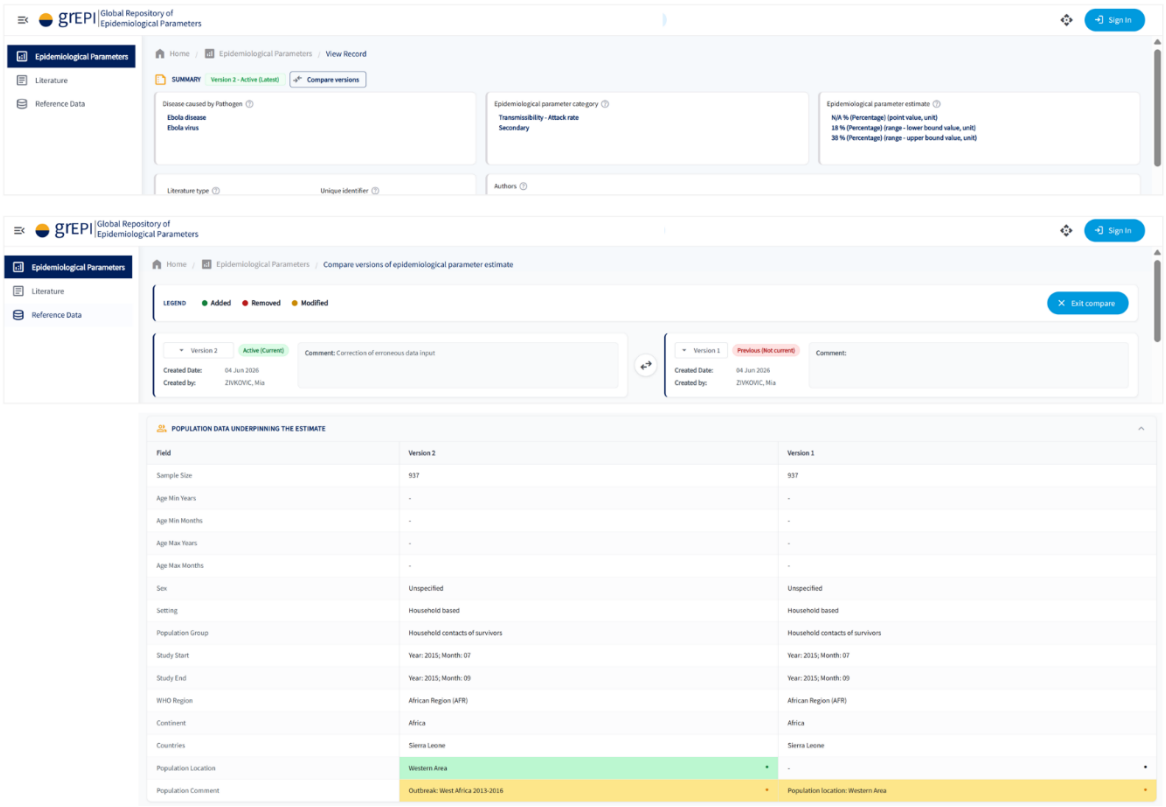



*Figure 9. The parameter detail view for a single estimate, showing the Compare version features*

# 4. Literature

The Literature section lists the source publications from which parameter estimates have been extracted.

**Default visible columns:** Literature type, Literature unique ID type, Unique identifier, Title, Authors, Published in (journal name), Publication Year, Last Modified Date.

**Additional columns available via Select:** Article ID, Paper copy only?, Publication volume, Publication issue, Page first, Page last, Label (short citation), Publication source type, Created date , Record Status.

The browse interface works that same way as in the Epidemiological Parameters table (see Section 3.1 for details on search, column filters, sorting, pagination, and export).

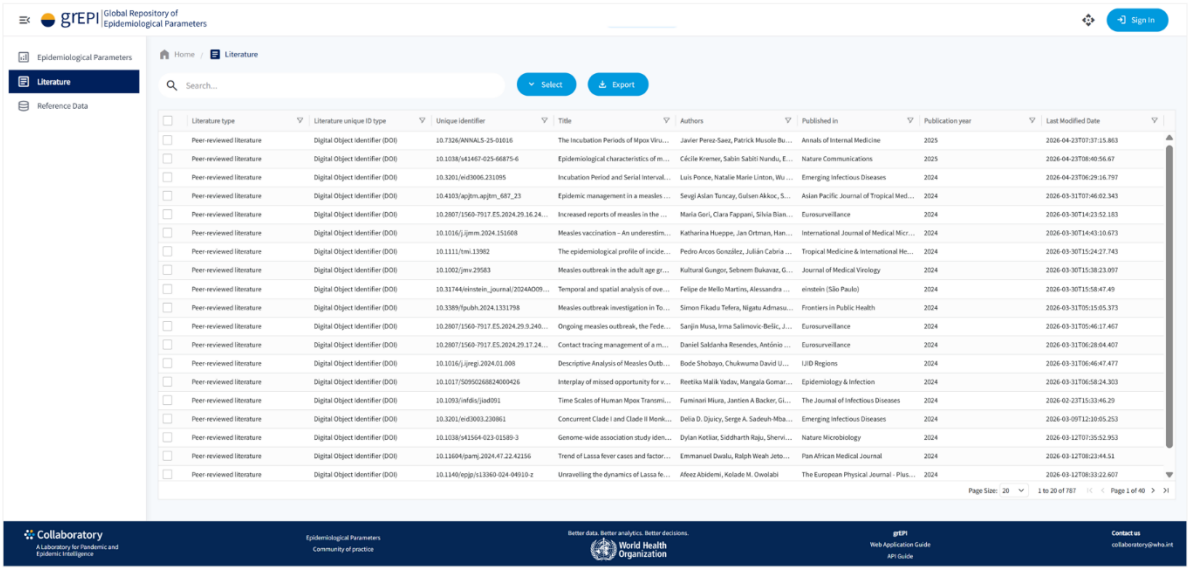



*Figure 10. Literature browse table – default view*

Each literature entry is linked to one or more parameter estimates. The Unique identifier serves as the key link between a parameter estimate and its source publication: the Literature unique identifier field in the Epidemiological Parameters table corresponds to the Unique identifier field in the Literature table.

**Change history.** The Action column, available only for signed in users, contains a three-dot menu for each row. Click it to access the History option, which opens a popup showing the change history for that entry. The History table displays the Field that was changed, the Date Time of the change, the User who made it, the Action Type (e.g., "Add new value"), and the Previous Value (if applicable). This provides a transparent audit trail for every literature record in grEPI.

**Copying values.** Values displayed in the browse tables can be copied directly from the grid without opening the detail view. To copy a value, click and hold the left mouse button to

highlight the desired text, then right-click and select “copy” from the context menu. This functionality is available in the Epidemiological Parameters, Literature, and Reference Data browse tables.

# 5. Reference Data

Reference Data holds the controlled vocabularies that define the valid values used throughout grEPI. The Reference Data landing page displays a card-based overview of all 35 reference tables, with a search bar to find a specific table by name. Click any card to open that reference table.

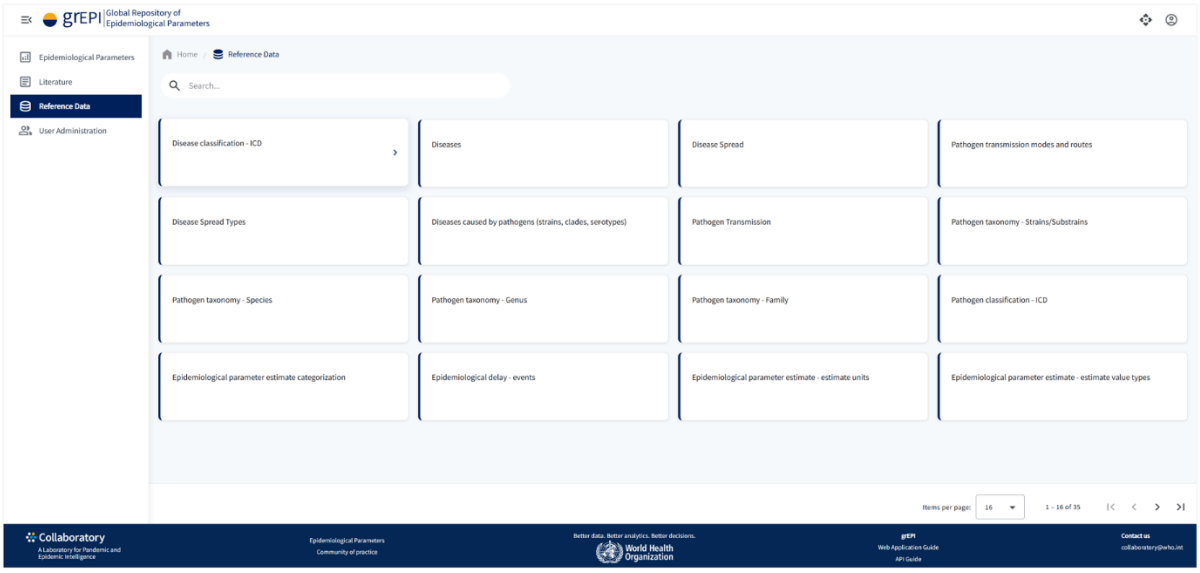



*Figure 11. The Reference Data landing page, showing the card-based grid of reference tables with the search bar at the top.*

The 35 reference tables fall into five groups. The Appendix provides a full column listing for each table.

**Disease and Pathogen Classification (12 tables):** Disease classification - ICD, Diseases, Disease Spread, Disease Spread Types, Diseases caused by Pathogens Pathogen classification - ICD, Pathogen taxonomy - Species, Pathogen taxonomy - Genus, Pathogen taxonomy - Family, Pathogen taxonomy - Strains/Substrains, Pathogen transmission modes and routes, Pathogen Transmission.

**Epidemiological Parameter Classification (8 tables):** Epidemiological parameter classification, Epidemiological delay - events, Epi parameter estimate - estimate units, Epi parameter estimate - estimate value types, Epi parameter estimate - uncertainty units, Epi parameter estimate - uncertainty value types, Mapping: categories to Percentage/Reproduction/Epi delay, Mapping: categories to allowed estimate units.

**Methods and Calculations (7 tables):** Distribution types, Probability distribution - Parameter value types, Methods to calculate reproduction number, Methods to calculate percentage, Percent Adjusted By and Data Disaggregated by - Tag mapping, Inference Methods, Epidemiological parameter estimates – Statistical approaches.

**Population and Study Context (5 tables):** Timing related to outbreaks - moment values, Population - Settings, Population - Groups, Population - Sex, Countries, territories, areas - WHO health boundaries.

**Literature (3 tables):** Literature types, Literature Unique Identifier types, Literature - Publication sources.

Each reference table has the same browse interface: search bar, Select (to choose visible columns), Export, column filters, sorting, and pagination.

**Change history.** Each row in a Reference Data table has a three-dot action menu in the Action column with a History option. This works the same way as in the Literature table: it shows a log of changes including the field, timestamp, user, action type, and previous value. For reference data entries created during initial system setup, the user may appear as "grepi_system_user". Currently, only designated administrators can modify reference data and literature records directly. Contributor features that allow registered users to suggest modifications are planned for a future release.

**Copying values.** Values displayed in the browse tables can be copied directly from the grid without opening the detail view. To copy a value, click and hold the left mouse button to highlight the desired text, then right-click and select “copy” from the context menu. This functionality is available in the Epidemiological Parameters, Literature, and Reference Data browse tables.

# 6. Exporting Data

Each of the three main sections (Epidemiological Parameters, Literature, and individual Reference Data tables) supports data export in CSV format.

## 6.1 What Gets Exported

**Columns.** The exported CSV includes only the columns that are currently selected via the Select button (see Section 3.1). To include all available fields in the export, open the Select dropdown and click "Select All" before exporting.

**Rows.** By default, the export includes all rows in the table (unfiltered). The export will be filtered if you have entered text in the search bar (only matching rows are exported) or if you have selected specific rows using the checkboxes (only those selected rows are exported).

## 6.2 How to Export

1. Navigate to the table you want to export.

2. Use the Select button to choose which columns to include. Click "Select All" if you want every available field.

3. Optionally, use the search bar to narrow the rows. For example, entering "measles incubation period" will limit the export to matching rows. Alternatively, use the checkboxes to select specific rows.

4. Click the Export button to download the CSV file.

# 7. API Access

grEPI data is also available in machine-readable format via the grEPI public API. The API enables programmatic access to epidemiological parameter estimates, literature records, and reference data. Two resources are available to support API usage:

The grEPI public **API endpoints(Swagger UI)** provides also the full technical specification of available endpoints, request parameters, and response schemas. It is accessible from the diamond-shaped icon in the top-right corner of the application, or directly at: https://collaboratory.who.int/grepi/api/swagger/index.html

The **grEPI API User Guide** provides guidance for working with the API, including quickstart examples in both R and Python, worked examples for common queries, and instructions for using grEPI data with the {epiparameter} R package. It is available at: https://who-collaboratory.github.io/collaboratory-grepi-web/assets/files/grEPI_API_Documentation.pdf

# 8. Frequently Asked Questions

**Why are some columns hidden by default?**

To keep the browse view readable, only the most commonly used fields are shown. The Epidemiological Parameters table has over 100 available fields. Click the Select button above any table to add or remove columns. You can search for column names within the Select dropdown.

**How do I export all fields, not just the visible ones?**

Click Select, then "Select All" to include all columns. Then click Export. If the search bar is empty and no rows are individually selected, the export will contain the full unfiltered dataset.

**Can I reset my custom selection of columns to the default column view in one go?**

Yes, by clicking **Reset** button which will appear next to the Export button in the browse form when at least one custom column has been added in the view.

**How does the search bar work?**

The search bar accepts multiple keywords separated by spaces. It matches each keyword against any field in the table and returns rows that satisfy all of the terms. For example, "mpox incubation period germany" will find rows where the disease is Mpox, the parameter subtype is Incubation period, and the country includes Germany.

**What filter options are available on columns?**

Click the filter icon on any column header to access these operators: Contains, Does not contain, Equals, Does not equal, Begins with, Ends with, Blank, and Not blank. Multiple column filters can be active at the same time.

**Can I clear all filters in one go?**

Yes, by clicking **Clear** button which will appear next to the Export button in the browse form when at least one column filter is applied in the table.

**What does the eye icon do?**

In the Epidemiological Parameters table, it opens the full detail view for a single parameter estimate, showing all fields organized into sections (Summary, Disease and Pathogen, Classification, Estimate, Probability Distribution, Methods, Population Data, Literature, and Data Sources).

**How do I find all parameter estimates from a specific study?**

Search or filter by the Literature unique identifier in the Epidemiological Parameters table. Alternatively, find the study in the Literature section and note its unique identifier, then filter the parameters table by that Unique identifier.

**How do I see previous versions of parameter estimate?**

Open the parameter detail view (click the eye icon in the Action column on the browse table). When a record has more than one version, a "Compare versions" option opens a side-by-side view that highlights differences between any two versions.

**Why does the browse table only show one row per estimate, even if there are multiple versions?**

The Epidemiological Parameters browse table shows the latest version of each estimate. This keeps the table focused on current values. Earlier versions remain available through the version history in the parameter detail view (see Section 3.3).

**What is the difference between Parameter Type and Parameter Subtype?**

The Type is the broad classification (e.g., Human Delay, Reproduction number, Case Fatality Ratio). The Subtype is the specific epidemiological parameter within that broader class. For example, under Human Delay: Incubation period, Serial interval, Generation interval. Under Reproduction number: Basic (R0), Effective (Re), Maximum (Rt).

**What do the "Data imported from" fields mean?**

These fields record where the data originally came from before being loaded into grEPI. Many of the current estimates were originally curated by partner initiatives (e.g, EpiReview/PERG at Imperial College London) and then imported into grEPI.

**How can I see when a record was last changed and by whom?**

In the Literature and Reference Data tables, click the three-dot menu in the Action column and select "History". This shows log of all changes to that record, including the date, the user who made the change, the type of action, and the previous value of any modified field. For parameter estimates, change tracking works differently: each update creates a new version, and the full version history is available from the parameter detail view (see Section 3.3).

**What does the question mark icon next to field names mean?**

These are in-app help tooltips. Hover over the circled question mark to see a brief description of what the field captures.

**Can I contribute data to grEPI?**

To get involved, contact the team via collaboratory@who.int or join the Epidemiological Parameters Community of Practice at https://collaboratory.who.int/about/communities/epidemiological-parameters/.

# Appendix: Reference Data Table Column Listings

This appendix lists every column available in each of the 35 Reference Data tables. All tables include standard audit fields (Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status) unless otherwise noted.

## Disease and Pathogen Classification

| Reference Table | Description |
|---|---|
| Disease classification - ICD | ICD chapter and sub-chapter classification codes |
| Diseases | Diseases covered by grEPI with ICD codes |
| Disease Spread | Disease-level transmission mode, route, and spread type mappings |
| Disease Spread Types | Types of disease spread (e.g., direct, indirect) |
| Diseases caused by Pathogens | Maps pathogen variants to diseases |
| Pathogen classification - ICD | Top-level ICD pathogen type classification |
| Pathogen taxonomy - Species | Pathogen species with ICD codes and genus linkage |
| Pathogen taxonomy - Genus | Pathogen genera with ICD codes and family linkage |
| Pathogen taxonomy - Family | Pathogen families with ICD codes and type linkage |
| Pathogen taxonomy - Strains/Substrains | Strains and substrains with full taxonomy hierarchy |
| Pathogen transmission modes and routes | Transmission modes and routes (e.g., airborne, vector-borne) |
| Pathogen Transmission | Pathogen-specific transmission mode and route assignments |

### 1. Disease classification - ICD

Columns: Disease Chapter ICD, Disease Chapter Code ICD, Disease Chapter ID ICD, Disease Chapter Depth In Kind, Disease Subchapter ICD, Disease Subchapter ID ICD, Disease Subchapter Depth In Kind, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**2. Diseases**

Columns: Disease, Disease ICD, Disease Code ICD, Disease ID ICD, Disease Subchapter ICD, Disease Chapter ICD, Disease Chapter Code ICD, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**3. Disease Spread**

Columns: Disease, Disease Spread Type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**4. Disease Spread Types**

Columns: Disease Spread Type, Created By, Created Date Time, Modified By, Modified Date Time, Record Status

**5. Diseases caused by Pathogens**

Columns: Disease, Pathogen Species, Pathogen Strain, Pathogen Substrain, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**6. Pathogen classification - ICD**

Columns: Pathogen type, Pathogen Type Name ICD, Pathogen Type Code ICD, Pathogen Type ID ICD, Pathogen T Depth In Kind, Created By, Created Date Time, Last Modified By, Last Modified Date, Record Status

**7. Pathogen taxonomy - Family**

Columns: Pathogen Family, Pathogen Family Name ICD, Pathogen Family Code ICD, Pathogen Family ID ICD, Pathogen Family Depth In Kind, Pathogen Type, Pathogen Type Name ICD, Pathogen Type Code ICD, Pathogen Type ID ICD, Pathogen Type Depth In Kind, Created By, Created Date Time, Last Modified By, Last Modified Date, Record Status

**8. Pathogen taxonomy - Genus**

Columns: Pathogen Genus, Pathogen Genus Name ICD, Pathogen Genus Code ICD, Pathogen Genus ID ICD, Pathogen Genus Depth In Kind, Pathogen Family, Pathogen Family Name ICD, Pathogen Family Code ICD, Pathogen Family ID ICD, Pathogen Family Depth In Kind, Created By, Created Date Time, Last Modified By, Last Modified Date, Record Status

**9. Pathogen taxonomy - Species**

Columns: Pathogen Species, Pathogen Species Name ICD, Pathogen Species Code ICD, Pathogen Species ID ICD, Pathogen Species Depth In Kind, Pathogen Genus, Pathogen Genus Name ICD, Pathogen Genus Code ICD, Pathogen Genus ID ICD, Pathogen Genus Depth In Kind, Created By, Created Date Time, Last Modified By, Last Modified Date, Record Status

**10. Pathogen taxonomy - Strains/Substrains**

Columns: Pathogen Strain, Pathogen Strain Name ICD, Pathogen Strain Code ICD, Pathogen Strain ID ICD, Pathogen Strain Depth In Kind, Pathogen Substrain, Pathogen Substrain Name ICD, Pathogen Substrain Code ICD, Pathogen Substrain ID ICD, Pathogen Substrain Depth In Kind, Pathogen Species, Pathogen Genus, Pathogen Family, Pathogen Type, Created By, Created Date Time, Last Modified By, Last Modified Date, Record Status

**11. Pathogen transmission modes and routes**

Columns: Pathogen Transmission Mode, Pathogen Transmission Route, Created By, Created Date Time, Last Modified By, Last Modified Date, Record Status

**12. Pathogen Transmission**

Columns: Pathogen Species, Pathogen Strain, Pathogen Substrain, Transmission Mode, Transmission Route, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

## Epidemiological Parameter Classification

| Reference Table | Description |
|---|---|
| Epidemiological parameter classification | Parameter classes, types, and subtypes (e.g., Delays > Human Delay > Incubation period) |
| Epidemiological delay - events | Events used in delay definitions (e.g., Infection, Symptom Onset, Death) |
| Epi parameter estimate - estimate units | Units for estimates (e.g., Days, Percentage) |
| Epi parameter estimate - estimate value types | How the estimate is expressed (e.g., Mean, Median) |

| | |
|---|---|
| Epi parameter estimate - uncertainty units | Units for uncertainty measures |
| Epi parameter estimate - uncertainty value types | Types of uncertainty values (e.g., CI, CrI) |
| Mapping: categories to Percentage/Reproduction/Epi delay | Flags which parameter subtypes are percentage-, reproduction-, or delay-related |
| Mapping: categories to allowed estimate units | Defines which units are valid for each parameter type and subtype |

**13. Epidemiological parameter classification**

Columns: Parameter Class, Parameter Type, Parameter Subtype, Created by, Created Date Time, Modified by, Modified Date Time, Record Status

**14. Epidemiological delay - events**

Columns: Epidemiological Event, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**15. Epi parameter estimate - estimate unit**

Columns: Estimate Unit, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**16. Epi parameter estimate - estimate value types**

Columns: Estimate Value Type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**17. Epi parameter estimate - uncertainty units**

Columns: Estimate Uncertainty Unit, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**18. Epi parameter estimate - uncertainty value types**

Columns: Estimate Uncertainty Value Type, Single Value Type, Paired/Range Value Type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 19. Mapping: categories to Percentage/Reproduction/Epi delay

Columns: Parameter Type, Parameter Subtype, Percentage related, Reproduction related, Epi Delay Related, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 20. Mapping: categories to allowed estimate unit

Columns: Parameter Type, Parameter Subtype, Estimate Unit, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

## Methods and Calculations

| Reference Table | Description |
|---|---|
| Distribution types | Types of fitted distributions (e.g., gamma, lognormal) |
| Probability distribution - Parameter value types | Types of distribution parameter values |
| Methods to calculate reproduction number | Methods for estimating R0, Re, Rt |
| Methods to calculate percentage | Methods for percentage-based estimates |
| Percent Adjusted By and Data Disaggregated by - Tag mapping | Dimensions/conditions that are used to adjust percent calculation or for data disaggregation |
| Epidemiological parameters - Inference Methods | Statistical inference methods used to derive estimates |
| Epidemiological parameters – Statistical approaches | Statistical approach used to estimate an epidemiological parameter |

### 21. Probability distribution types

Columns: Distribution Type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 22. Probability distribution - Parameter value types

Columns: Probability Distribution Parameter Value Type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 23. Methods to calculate reproduction number

Columns: Estimate Method Reproduction Number, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 24. Methods to calculate percentage

Columns: Percent Calculation, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 25. Percent Adjusted By and Data Disaggregated by - Tag mapping

Columns: Tag, Used for Percent Adjusted By?, Used for Data Disaggregation, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 26. Epidemiological parameters - Inference Methods

Columns: Inference Method, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 27. Epidemiological parameters – Statistical approaches

Columns: Statistical approach, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

## Population and Study Context

| Reference Table | Description |
|---|---|
| Timing related to outbreaks - moment values | When data was collected relative to the outbreak |
| Population - Settings | Study settings (e.g., hospital-based, community) |
| Population - Groups | Population groups studied |
| Population - Sex | Sex categories for the study population |
| Countries, territories, areas - WHO health boundaries | Geographic reference with ISO codes, WHO region, World Bank income group, and continent |

### 28. Timing related to outbreaks - moment values

Columns: Estimate Method Moment Value, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status. Note: the CSV export filename for this table currently reads "Epidemiological Parameter Estimate Method Moment Value"; this will be corrected in a future release.

### 29. Population - Settings

Columns: Population Setting, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 30. Population - Groups

Columns: Population Group, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 31. Population - Sex

Columns: Population Sex, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

### 32. Countries, territories, areas - WHO health boundaries

Columns: Country Name, ISO2 Country Code, ISO3 Country Code, WHO Region, World Bank Income Group, Continent, Created by

## Literature

| Reference Table | Description |
|---|---|
| Literature types | Types of literature (e.g., peer-reviewed, grey) |
| Literature Unique Identifier types | Types of literature unique identifiers (e.g., DOI, URL, PMCID) |
| Literature - Publication sources | Types of publication sources (e.g., Journal) |

### 33. Literature types

Columns: Literature type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**34. Literature Unique Identifier types**

Columns: Literature Unique ID type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status

**35. Literature - Publication sources**

Columns: Publication source type, Created by, Created Date Time, Last Modified By, Last Modified Date, Record Status